

Final Report

Nexus Weather & Climate

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Company:	Nexus Weather & Climate
Location:	Remote (primary) — Boulder, CO
Average Hours/Week:	10–15
Course Due Date:	December 5, 2025 (submit early)

Introduction

Nexus Weather & Climate builds machine learning calibrations that convert raw numerical weather model output into reliable guidance for decision makers. The company focuses on post-processing ensemble forecasts to reduce bias and variance at the surface variable level. As an intern I work on training and validating regression models that improve forecasts for core and precipitation-related targets. Our pipeline emphasizes reproducibility, automated evaluation, and clear logging so results are easy to compare across data drops. Work happens primarily in Python with common libraries such as pandas, NumPy, scikit-learn, XGBoost, and Matplotlib. Collaboration is remote and centered on Slack and GitHub with frequent short updates. During Fall 2025 the dataset expanded to include additional 2024 observations and two evaluation modes: in-window and out-of-window. Those additions improved stability in many targets while showing where more data is still needed. The internship began on August 25, 2025. This paper summarizes goals, weekly work, methods, teamwork, reflections, and outcomes.

The forecasting problem is naturally noisy and highly seasonal, which makes generalization across dates challenging. To manage this, the team uses consistent train and test splits, a shared evaluation harness, and standardized output folders. Each run records mean absolute error, mean squared error, and pre/post-ML scatter plots for quick visual checks. Inputs are validated with simple gates so inference only proceeds when coordinates and feature counts match expectations. This prevents misleading results and saves time in debugging. The approach is practical: start with strong baselines, measure carefully, and make incremental changes that are easy to adopt. The work favors clarity over complexity and documents both wins and limits.

A key theme this term was reducing turnaround time while keeping results trustworthy. Partial training was introduced so small configuration changes no longer force a full retrain. In parallel, loss functions were reviewed to handle outliers that often appear in precipitation targets. Huber loss, which blends squared and absolute error, proved more stable under extremes while preserving good behavior for the bulk of the distribution. Together these two upgrades shortened iteration cycles and reduced spiky outcomes. The changes were small in code but meaningful in day-to-day use. Decisions were based on side-by-side comparisons, error breakdowns, and simple parity checks.

Another theme was organization. Artifacts are stored under a clear **Output/** hierarchy with per-run READMEs. The folder layout includes models, tables, and plots so any result can be reproduced or compared later. A lightweight run registry captures which dataset, split, features, loss, and seed were used. This makes it straightforward to answer questions like why two figures differ or which settings led to a given improvement. Clean records also help the team collaborate because each person can rely on the same checklist and naming rules.

Overall, the internship provided hands-on experience with applied forecasting, careful evaluation, and practical engineering. The scope covered baselines, validation logic, resume training, loss selection, and result organization. The work built on summer progress and advanced toward production-friendly practices. The following sections present the weekly journal, detailed data science experience, notes about the team, reflections on changes, likes, and dislikes, and concluding thoughts.

Weekly Journal (Goal 1)

Week 1 (Aug 25 – Aug 31) — ~12 hours

Dates & Hours Worked

Aug 25 – Aug 31, ~12 hours

Team Members You Worked With

Luke Peffers (CEO/Manager); Simran Jadhav; Atharv

Project/Task Description

Repo pull and environment validation in Python; ensured consistent **Output/** schema. Helped Atharv set up credentials, repo access, venv/conda, and the run ledger; walked him through training and inference flow. Tools: Python, scikit-learn, pandas.

Key Collaboration

Simran and Atharv for onboarding and checks; Luke for expectations and directory schema alignment.

Outcome/Deliverables

Onboarding checklist; baseline run instructions; artifact folder template and README.

Enjoyable Aspects

Collaborative setup and clear ownership from day one.

Challenges/Dislikes

Reconciling historical variable-set differences across saved models.

Week 2 (Sep 1 – Sep 7) — ~13 hours

Dates & Hours Worked

Sep 1 – Sep 7, ~13 hours

Team Members You Worked With

Luke Peffers (CEO/Manager); Simran Jadhav; Atharv

Project/Task Description

Re-ran baselines (Random Forest, GBRT, XGBoost, SVR) on refreshed drops using a shared MAE/MSE evaluation function and automated pre and post ML plots. Tools: Python, scikit-learn, xgboost, matplotlib.

Key Collaboration

Luke for evaluation alignment; Simran and Atharv for pair checks on outputs and plot exports.

Outcome/Deliverables

Saved models per target; consolidated MAE/MSE tables; pre and post scatter plots.

Enjoyable Aspects

Quantifiable gains over raw forecasts; smooth teamwork.

Challenges/Dislikes

Enforcing fair, comparable splits across all runs.

Week 3 (Sep 8 – Sep 14) — ~12 hours

Dates & Hours Worked

Sep 8 – Sep 14, ~12 hours

Team Members You Worked With

Luke Peffers (CEO/Manager); Simran Jadhav; Atharv

Project/Task Description

Strengthened inference gating: lat/lon conformity, feature-count checks, and pre/post sanity thresholds required before allowing inference. Atharv contributed test cases; Simran reviewed logs. Tools: Python, numpy, pandas.

Key Collaboration

Atharv for test scenarios; Simran for log review; Luke for threshold guidance.

Outcome/Deliverables

Stricter validation gates; clearer rejection messages and logs.

Enjoyable Aspects

Turning hidden issues into explicit, actionable checks.

Challenges/Dislikes

Handling edge cases with only marginal post-ML gains.

Week 4 (Sep 15 – Sep 21) — ~11 hours

Dates & Hours Worked

Sep 15 – Sep 21, ~11 hours

Team Members You Worked With

Luke Peffers (CEO/Manager); Simran Jadhav; Atharv

Project/Task Description

Split-size sensitivity (in-window). Documented variance across models and targets due to data volume; Atharv replicated on a second variable set. Tools: Python, scikit-learn, matplotlib for sensitivity plots.

Key Collaboration

Luke for experiment focus; Simran and Atharv for parallel runs.

Outcome/Deliverables

Sensitivity charts and a short memo summarizing split-induced variance.

Enjoyable Aspects

Seeing how split choices affect generalization.

Challenges/Dislikes

Data scarcity magnifying variance.

Week 5 (Sep 22 – Sep 28) — ~14 hours

Dates & Hours Worked

Sep 22 – Sep 28, ~14 hours

Team Members You Worked With

Luke Peffers (CEO/Manager); Simran Jadhav; Atharv

Project/Task Description

Pulled `luke-new-ensemble-structure` branch; adopted 5 percent test split; added in-window and out-of-window evaluation per CEO guidance on Fri, Sep 26. Observed out-of-window sensitivity and improved stability with 2024 data. Tools: Python, scikit-learn, xgboost, matplotlib.

Key Collaboration

Luke for experimentation guidance during backfill planning.

Outcome/Deliverables

Side-by-side in-window and out-of-window plots and MAE tables across targets.

Enjoyable Aspects

Immediate feedback from broader temporal coverage.

Challenges/Dislikes

Clear signals of data deprivation out-of-window.

Week 6 (Sep 29 – Oct 5) — ~13 hours**Dates & Hours Worked**

Sep 29 – Oct 5, ~13 hours

Team Members You Worked With

Luke Peffers (CEO/Manager); Simran Jadhav; Atharv

Project/Task Description

Tried alternative regressors with minimal tuning under constraints. Standardized `Output/` folders and per-run READMEs; maintained a run ledger for reproducibility. Tools: Python, scikit-learn, pandas, matplotlib.

Key Collaboration

Simran and Atharv to avoid experiment overlap and share best settings per target family.

Outcome/Deliverables

Consolidated results table; refreshed plots; updated run README and folder schema.

Enjoyable Aspects

Methodical iteration with a clear division of focus.

Challenges/Dislikes

Compute and time trade-offs limiting broader hyperparameter searches.

Week 7 (Oct 6 – Oct 12) — ~12 hours**Dates & Hours Worked**

Oct 6 – Oct 12, ~12 hours

Team Members You Worked With

Luke Peffers (CEO/Manager); Simran Jadhav; Atharv

Project/Task Description

Led the XGBoost resume path so training can continue from a saved model instead of starting

over. Set the file naming pattern and tracked training iterations so resumes pick up cleanly. Simran focused on GBRT warm start, and Atharv drafted the safety checklist and helper.

Key Collaboration

Simran on GBRT warm start handoff; Atharv on safety checks; Luke for scope approval.

Outcome/Deliverables

Resume workflow for XGBoost, naming convention, and iteration tracking notes.

Enjoyable Aspects

Seeing resume load and continue exactly where a prior run stopped.

Challenges/Dislikes

Parameter changes that break resume and force a clean retrain.

Week 8 (Oct 13 – Oct 19) — ~12 hours

Dates & Hours Worked

Oct 13 – Oct 19, ~12 hours

Team Members You Worked With

Luke Peffers (CEO/Manager); Simran Jadhav; Atharv

Project/Task Description

Integrated a preflight validator and a metadata schema: each saved model now records split, features, seed, and estimator count, and the validator blocks unsafe resumes. Simran wired GBRT warm start. Atharv ran negative tests to prove the validator catches mismatches.

Key Collaboration

Simran for GBRT integration details; Atharv for negative-test coverage; Luke for accept criteria.

Outcome/Deliverables

Validator module, metadata schema, and parity checks showing resume vs full retrain match.

Enjoyable Aspects

Confidence that a resume only runs when it is safe to do so.

Challenges/Dislikes

Updating a few older artifacts that lacked complete metadata.

Week 9 (Oct 20 – Oct 26) — ~13 hours

Dates & Hours Worked

Oct 20 – Oct 26, ~13 hours

Team Members You Worked With

Luke Peffers (CEO/Manager); Simran Jadhav; Atharv

Project/Task Description

Owned the loss-function experiment plan and tuned the Huber threshold. Ran precipitation targets and built error-by-quantile charts to show stability under extremes. Simran handled core variables and plotting polish; Atharv summarized results into compact tables.

Key Collaboration

Simran for core-variable runs and figure cleanup; Atharv for tables; Luke for review of the plan.

Outcome/Deliverables

Experiment matrix, Huber selection notes, and quantile error visuals for precipitation.

Enjoyable Aspects

Robustness gains when outliers would otherwise skew results.

Challenges/Dislikes

One threshold cannot be perfect for every target.

Week 10 (Oct 27 – Nov 2) — ~12 hours**Dates & Hours Worked**

Oct 27 – Nov 2, ~12 hours

Team Members You Worked With

Luke Peffers (CEO/Manager); Simran Jadhav; Atharv

Project/Task Description

Updated configs and the evaluator to record loss choice and key parameters, and set the default policy: use Huber for skew-prone targets. Simran reproduced results across assets and refreshed docs; Atharv backfilled comparisons and ran sanity checks.

Key Collaboration

Simran for replication and docs; Atharv for backfill checks; Luke for policy sign off.

Outcome/Deliverables

Config updates, evaluator logging, and a simple defaulting rule for loss functions.

Enjoyable Aspects

Cleaner records that make comparisons straightforward.

Challenges/Dislikes

Care needed when comparing new Huber runs to older squared-error results.

Week 11 (Nov 3 – Nov 7) — ~11 hours**Dates & Hours Worked**

Nov 3 – Nov 7, ~11 hours

Team Members You Worked With

Luke Peffers (CEO/Manager); Simran Jadhav; Atharv

Project/Task Description

Wrote the implementation note and finalized defaults: Huber for precipitation and other skewed targets; squared error for near-normal ones. Organized **Output/** folders and a simple run registry so future comparisons are easy. Simran peer-reviewed guidelines; Atharv cleaned older artifacts and fixed missing metadata.

Key Collaboration

Luke for final approval; Simran for editorial review; Atharv for artifact cleanup.

Outcome/Deliverables

Implementation note, updated READMEs, organized artifacts, and a lightweight run registry.

Enjoyable Aspects

Closing the loop from idea to adopted default with clear traceability.

Challenges/Dislikes

Chasing small inconsistencies in older runs.

Data Science Experience

The work starts with a baseline suite: Random Forest, Gradient Boosted Trees, XGBoost, and SVR, all trained under the same protocol and evaluated with a shared utility. The harness computes MAE and MSE and exports pre and post ML scatter plots for each target. This consistency removes ambiguity when comparing models or new data drops. Inputs are checked before training and inference so coordinate ranges and feature counts are correct. Clear messages report why a run is blocked, which speeds debugging. These simple steps prevent many downstream errors.

Resume training was added to shorten iteration cycles. For XGBoost I implemented a resume path that continues training from a saved model, while Simran integrated warm start for GBRT. A small metadata record is saved with every model so a preflight validator can confirm settings like split, features, and seed. Atharv created a safety checklist and negative tests to prove the validator blocks unsafe resumes. When settings match, resume delivers the same accuracy as a clean retrain but in less time. When settings differ, the system forces a clean run.

Loss selection was revisited to handle extreme values. I designed experiments that compared squared error, absolute error, and Huber loss across precipitation targets with error-by-quantile charts. Huber provided steadier behavior when forecast errors contained outliers without hurting performance in typical ranges. Simran standardized figures for core variables so comparisons are uniform. Atharv summarized the outcomes into compact tables so results can be scanned quickly. The team agreed to make Huber the default for skew-prone targets and to keep squared error for near-normal ones.

Artifacts are organized so results remain useful later. Each run writes models, tables, and plots to a uniform `Output/` layout along with a short README. I added evaluator logging to record which loss and key parameters were used in each run. A simple run registry lists dataset name, split size, features, loss, and seed, which helps avoid duplicated work and keeps comparisons fair. This structure reduces rework and makes handoffs easier across the team.

The experience combined modeling with practical engineering: consistent evaluation, validation gates, resume logic, sensible loss choices, and disciplined organization. The gains are modest by design, but they accumulate and make the system faster, safer, and easier to maintain. Most importantly, every change is measurable and reversible, which keeps risk low while progress continues.

About the Team

I set priorities with Luke Peffers and work closely with Simran Jadhav and Atharv. We split ownership to avoid overlap and keep progress steady. I led the XGBoost resume path and loss-experiment design; Simran focused on GBRT warm start, standard figures, and docs; Atharv covered safety checks, negative tests, result tables, backfills, and artifact cleanup. We review plots and tables together in Slack and GitHub before adopting any change. The collaboration is straightforward and relies on small, verifiable steps.

Changes, Likes, and Dislikes

I would keep partial training and Huber defaults moving forward and add a lightweight internal page that lists datasets, splits, losses, seeds, and key metrics for each run. I liked the measurable improvements over raw guidance, the shorter cycle time after resume support, and the reduced noise on skewed targets with Huber loss. Challenges came from data scarcity in out-of-window testing and from careful comparisons with older squared-error runs. These issues are manageable with clean records and clear rules.

Conclusions

This internship strengthened practical skills for applied forecasting: build solid baselines, measure carefully, validate inputs, resume when safe, pick loss functions that match the error shape, and keep artifacts organized. The work delivered faster iteration and steadier results without adding unnecessary complexity. The habits developed here will transfer well to future projects where clarity and reliability matter as much as raw accuracy.